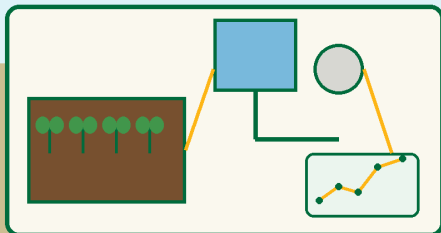


TRIAL OBJECTIVE

OBJECTIVE



OBJECTIVE

TRIAL OBJECTIVE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A crop trial evaluates a defined treatment, variety, practice, or system using planned comparisons and measurable outcomes.

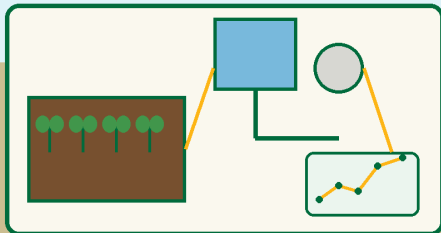
ANALYSIS / DESIGN TASK

Write a precise objective for a school-scale trial.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

NULL HYPOTHESIS

H0



H0

NULL HYPOTHESIS

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

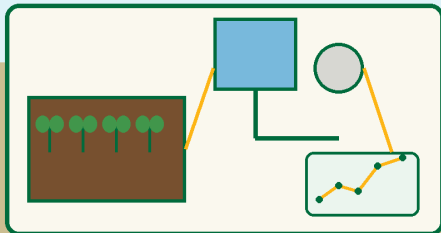
A null hypothesis states that there is no specified treatment effect or difference in the population being studied.

ANALYSIS / DESIGN TASK

Write H_0 for a two-treatment comparison.

ALTERNATIVE HYPOTHESIS

H1



H1

ALTERNATIVE HYPOTHESIS

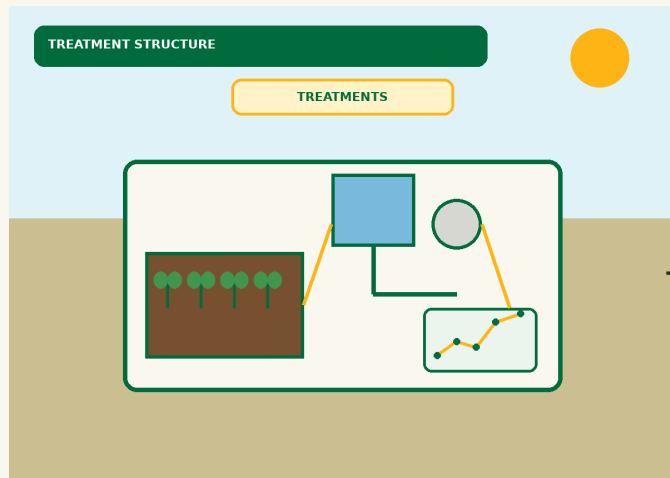
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

An alternative hypothesis states the treatment effect or difference that competes with the null hypothesis.

ANALYSIS / DESIGN TASK

Write a corresponding H1.



TREATMENTS

TREATMENT STRUCTURE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

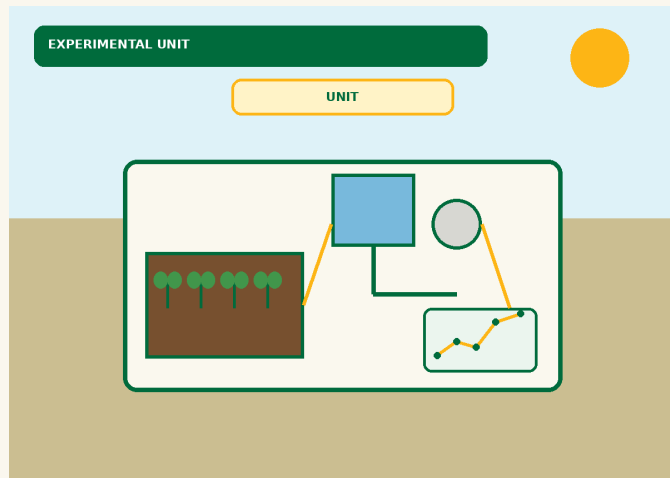
TECHNICAL CONCEPT

Treatment structure defines the factors, levels, and combinations being evaluated.

ANALYSIS / DESIGN TASK

Create a treatment table for one factor.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



UNIT

EXPERIMENTAL UNIT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

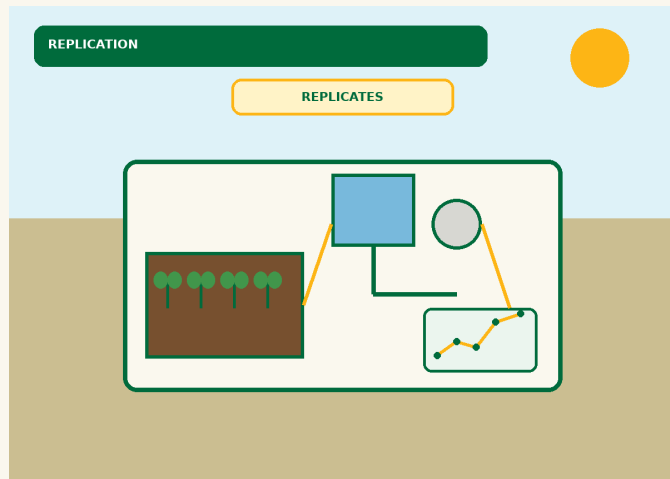
TECHNICAL CONCEPT

The experimental unit is the smallest unit independently receiving a treatment and determines valid replication.

ANALYSIS / DESIGN TASK

Identify the experimental unit and justify it.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



REPLICATES

REPLICATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

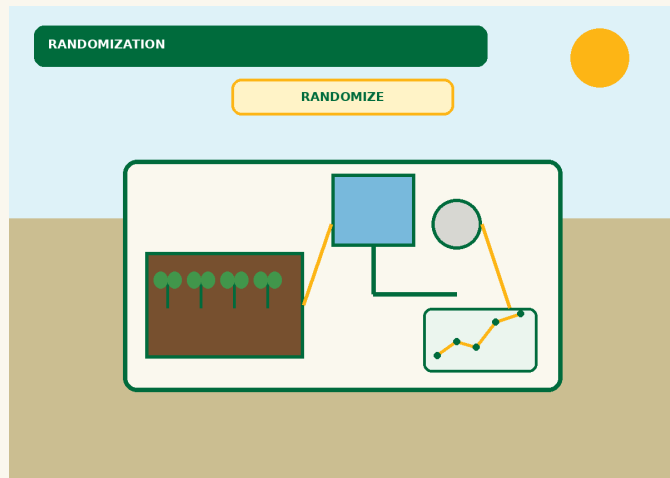
TECHNICAL CONCEPT

Replication provides independent observations of treatment conditions and supports estimation of experimental variation.

ANALYSIS / DESIGN TASK

Determine the number of true replicates.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



RANDOMIZE

RANDOMIZATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Randomization assigns treatments by chance to reduce systematic bias and support valid inference.

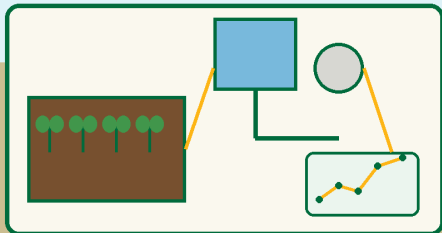
ANALYSIS / DESIGN TASK

Generate a random treatment allocation.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

BLOCK DESIGN

BLOCK



BLOCK

BLOCK DESIGN

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

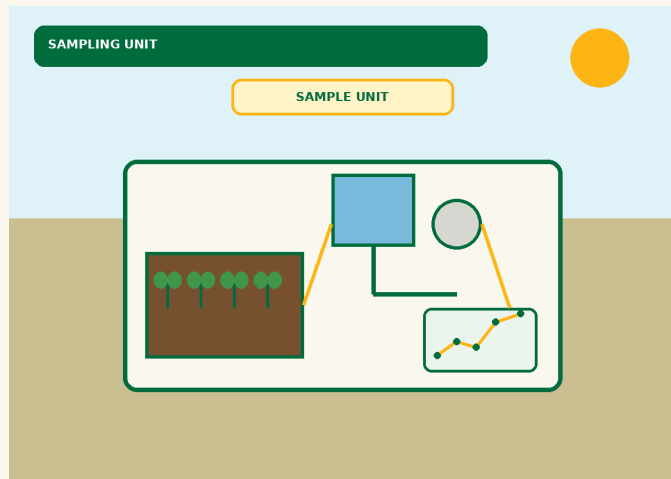
TECHNICAL CONCEPT

Blocking groups relatively similar experimental units so known spatial variation can be separated from treatment comparison.

ANALYSIS / DESIGN TASK

Design a simple randomized complete block layout.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



SAMPLE UNIT

SAMPLING UNIT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

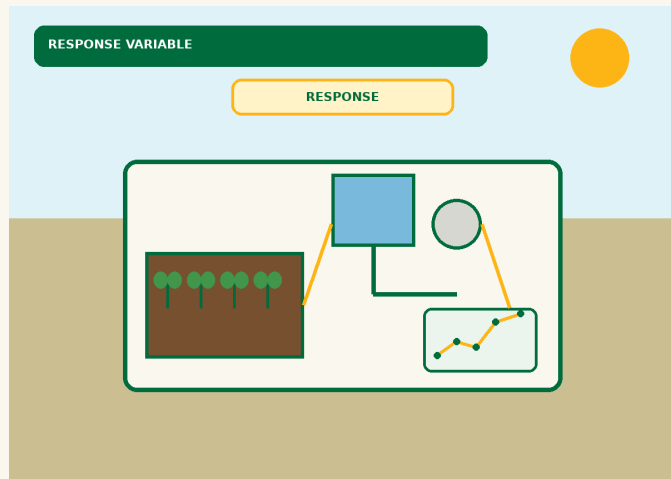
TECHNICAL CONCEPT

A sampling unit is the plant, quadrat, fruit, leaf, or other unit actually measured within an experimental unit.

ANALYSIS / DESIGN TASK

Distinguish sampling units from experimental units.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



RESPONSE

RESPONSE VARIABLE

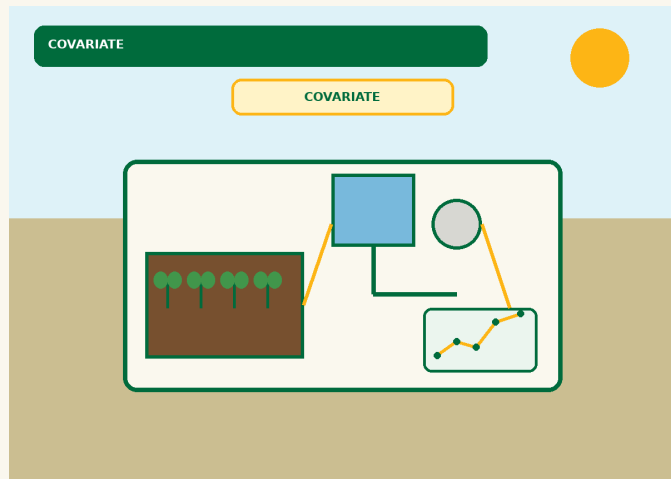
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A response variable is the measured outcome used to evaluate treatment effects.

ANALYSIS / DESIGN TASK

Define its unit and measurement timing.



COVARIATE

COVARIATE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A covariate is a measured variable not assigned as a treatment that may help explain response variation.

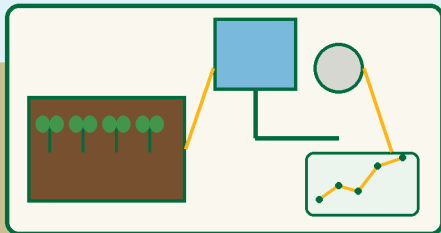
ANALYSIS / DESIGN TASK

Identify a useful baseline covariate.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

STANDARDIZED PROTOCOL

PROTOCOL



PROTOCOL

STANDARDIZED PROTOCOL

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Standardized protocols reduce measurement differences caused by inconsistent procedures, instruments, timing, or observers.

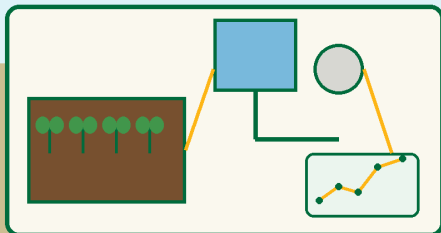
ANALYSIS / DESIGN TASK

Draft a concise measurement protocol.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

DATA DISTRIBUTION

DISTRIBUTION



DISTRIBUTION

DATA DISTRIBUTION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

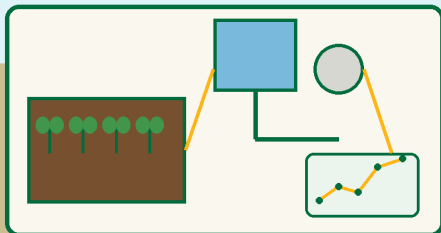
A data distribution describes the observed pattern, center, spread, and frequency of measured values.

ANALYSIS / DESIGN TASK

Inspect a small dataset for shape and spread.

MEAN & STANDARD DEVIATION

MEAN + SD



MEAN + SD

MEAN & STANDARD DEVIATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

CARD 834 • MEAN & STANDARD DEVIATION

TECHNICAL CONCEPT

The mean summarizes central tendency while standard deviation describes dispersion around the mean.

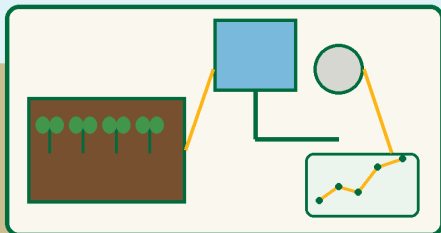
ANALYSIS / DESIGN TASK

Calculate both from teacher-provided data.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

STANDARD ERROR

SE



SE

STANDARD ERROR

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

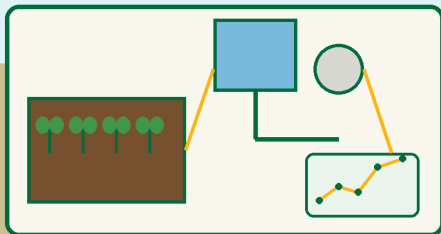
Standard error describes uncertainty in an estimated mean and depends on variability and sample size.

ANALYSIS / DESIGN TASK

Interpret provided means and standard errors.

CONFIDENCE INTERVAL

CI



CI

CONFIDENCE INTERVAL

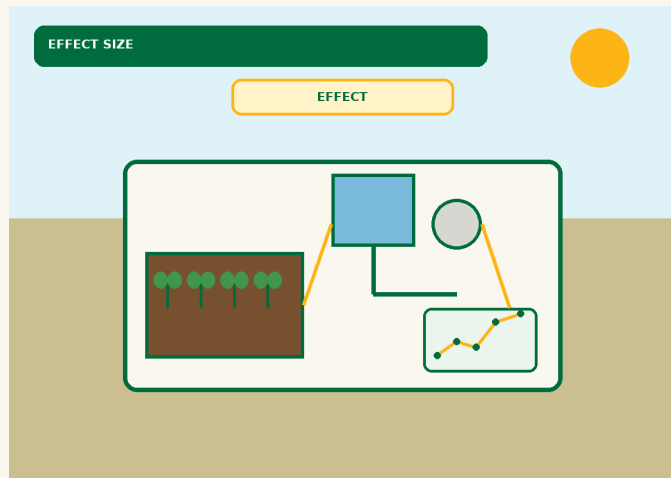
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A confidence interval gives a range of plausible values for an estimated population parameter under stated statistical assumptions.

ANALYSIS / DESIGN TASK

Interpret a provided confidence interval.



EFFECT

EFFECT SIZE

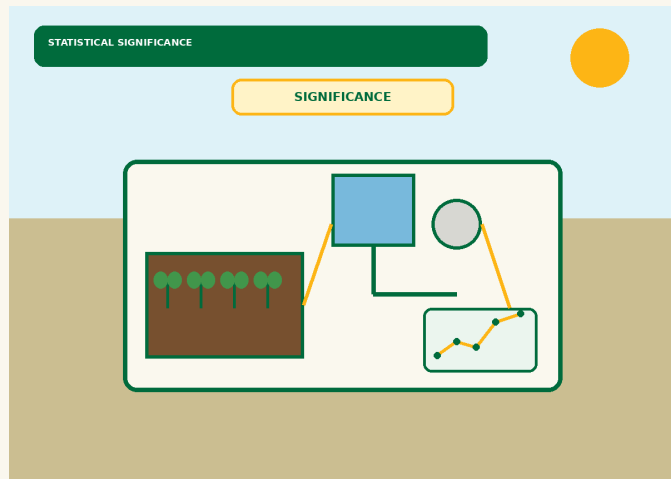
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Effect size quantifies the magnitude of a treatment difference or relationship rather than only whether it is statistically detectable.

ANALYSIS / DESIGN TASK

Calculate a simple absolute or relative effect.



SIGNIFICANCE

STATISTICAL SIGNIFICANCE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

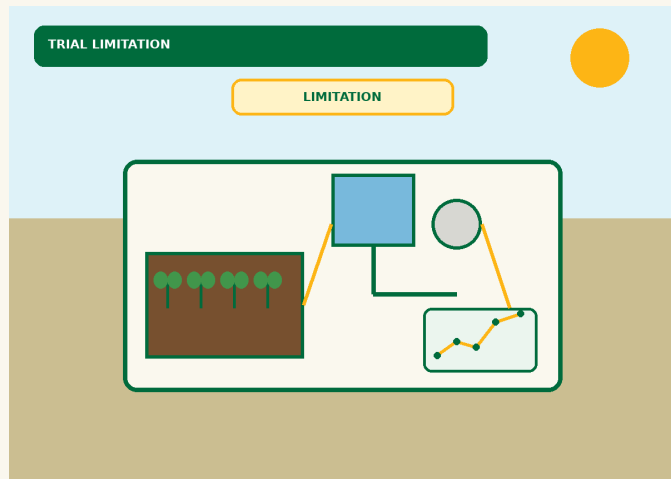
TECHNICAL CONCEPT

Statistical significance evaluates data against a specified statistical model and threshold; it does not by itself establish practical importance.

ANALYSIS / DESIGN TASK

Distinguish statistical and practical importance in a scenario.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



LIMITATION

TRIAL LIMITATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

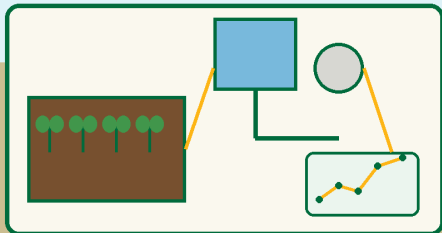
Trial limitations include design, sample size, measurement, duration, environmental variation, and scope constraints.

ANALYSIS / DESIGN TASK

Identify limitations that affect interpretation.

TRIAL RECOMMENDATION

RECOMMEND



RECOMMEND

TRIAL RECOMMENDATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A trial recommendation should reflect effect magnitude, uncertainty, feasibility, cost, and the conditions under which results were obtained.

ANALYSIS / DESIGN TASK

Write a cautious evidence-based recommendation.